Comparision benchmark of 1024 env GPU vs CPU

It was observed that CPU outperforms GPU at low environment counts (NUM_ENVS=256). However, as NUM_ENVS increases, CPU throughput saturates — limited by the number of physical cores — while GPU utilization increases proportionally with the number of parallel environments. This allows the GPU to exploit its thousands of CUDA cores, reducing total training time significantly at higher environment counts.

Duvvuri Venkata Naga Sai Ritvik me25b039

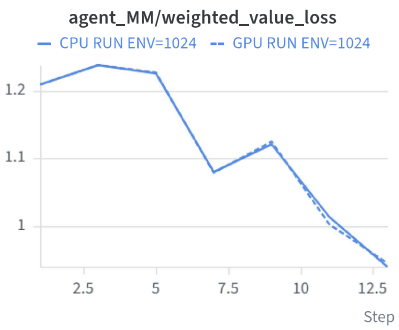
NOTE : All these Simulations were ran on a Laptop with intel i7-13700HX processor and Nvidia RTX 4060 GPU

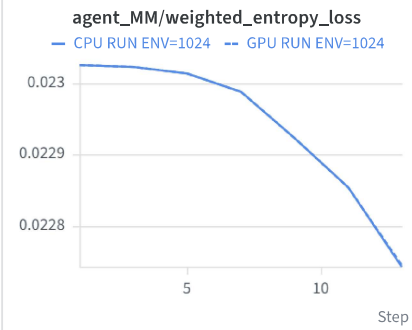
EXE Agent Metrics

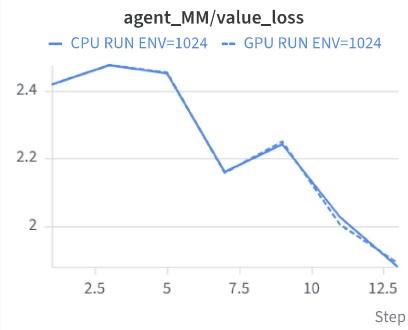


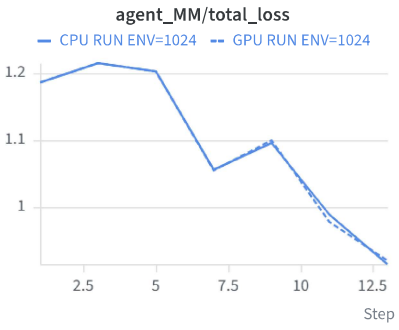
There are minor deviations Probably because of floating point Errors

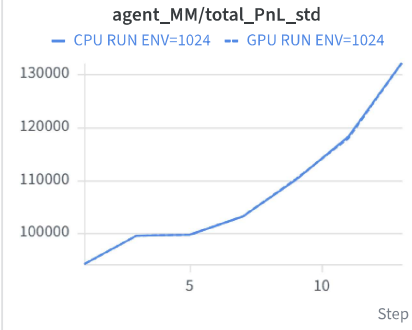
Add markdown, images, and $\LaTeX$

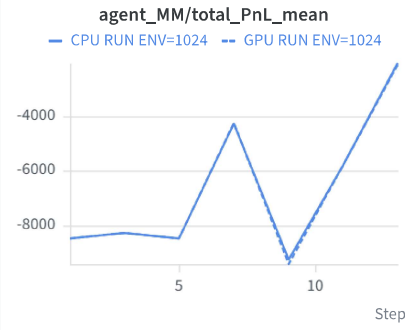




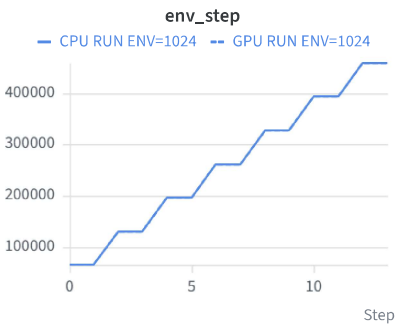








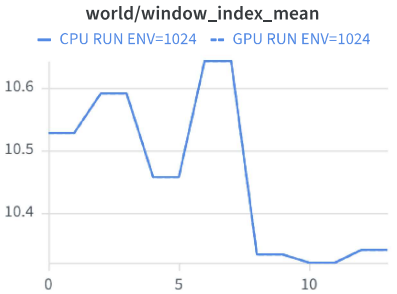
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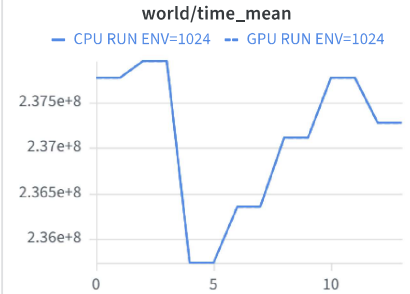


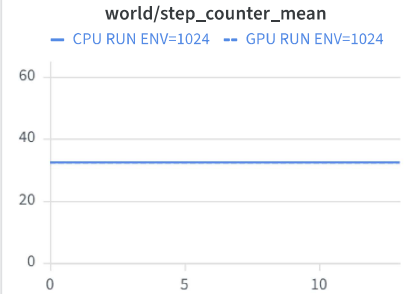
Add markdown, images, and $\LaTeX$

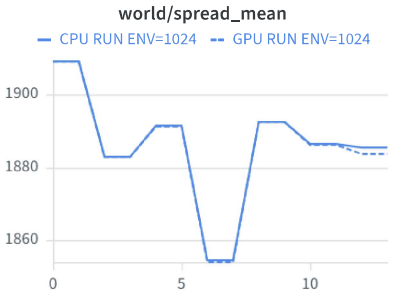
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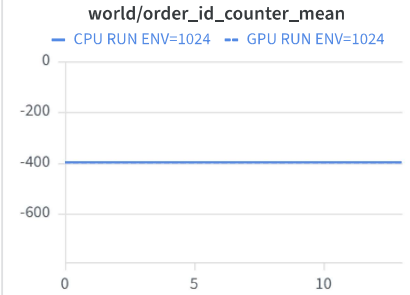
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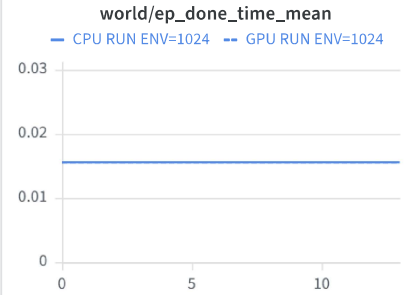






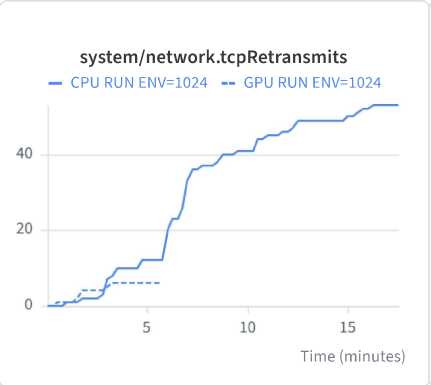


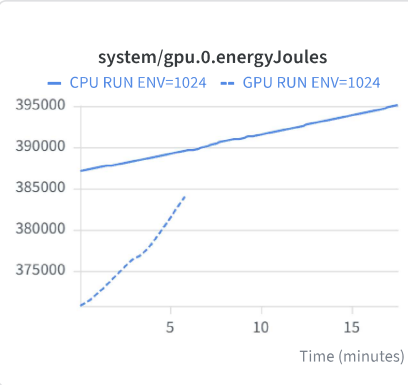


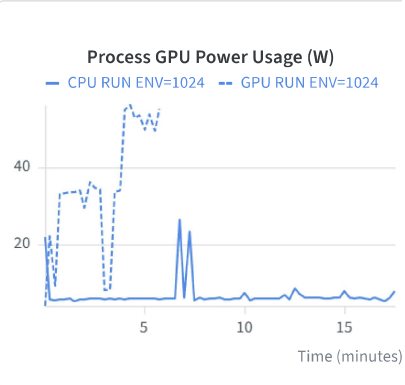


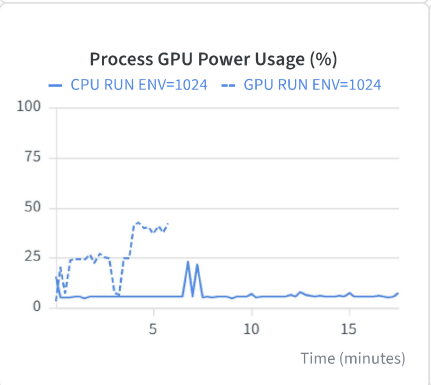
Environment-level metrics (step counter, episode done time, order counter) are perfectly identical across CPU and GPU runs, confirming simulation correctness. Market-level metrics (spread, time, window index) show minor divergence attributable to data sampling order differences from floating point non-determinism, with both runs converging to the same final values.

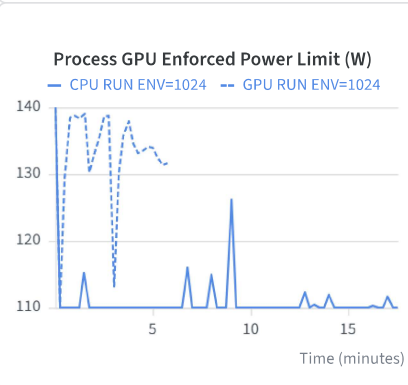
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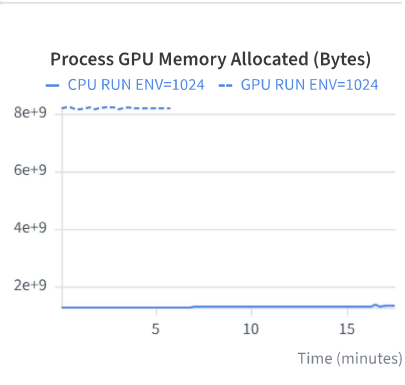






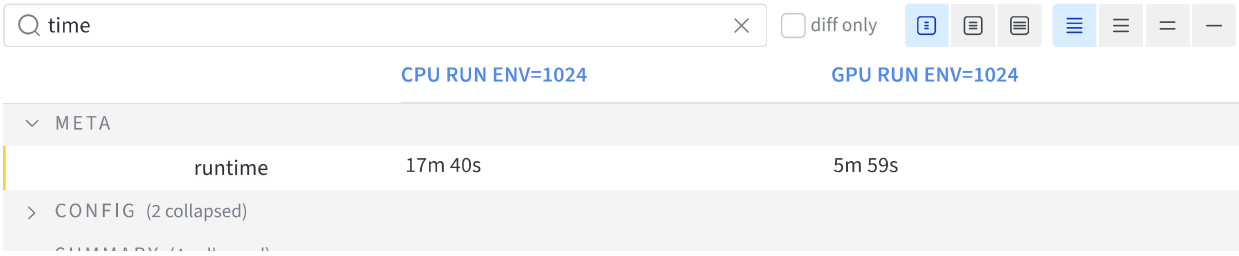






Key numbers

Metric	GPU Run	CPU Run
Wall time	~6 min	~17 min
Energy (J)	~384,000 (still rising)	~395,000 (still rising)
GPU power (W)	35–45 W (JIT spike)	~5 W (idle)
GPU power %	up to 40%	~5%
GPU memory	~8 GB	~1.1 GB



It is Clearly seen that run time is faster for GPU RUN ENV =1024 Than that of CPU's